Monetary Policy in Practice

CHAPTER 15 · PRINCIPLES OF MACROECONOMICS

Policy decisions and the long-run effects of monetary policy

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Learning Objectives

- Measure output gaps and evaluate policy responses to demand and supply shocks.
- Calculate and interpret a standard Taylor-rule benchmark.
- Compare a policy-rate rule with an inflation-targeting framework and explain their limits.
- Explain the long-run effects of monetary policy and monetary neutrality: why changes in the money supply affect the price level but not real output in the long run.

Choosing a Policy Response

- The Fed pursues maximum employment and price stability. Its longer-run inflation goal is 2% PCE inflation.
- Compare actual output with sustainable potential output; evaluate inflation and its expected path.
- A demand shortfall and a supply disruption can produce different policy tradeoffs.

Federal Reserve Board: policy goals and transmission

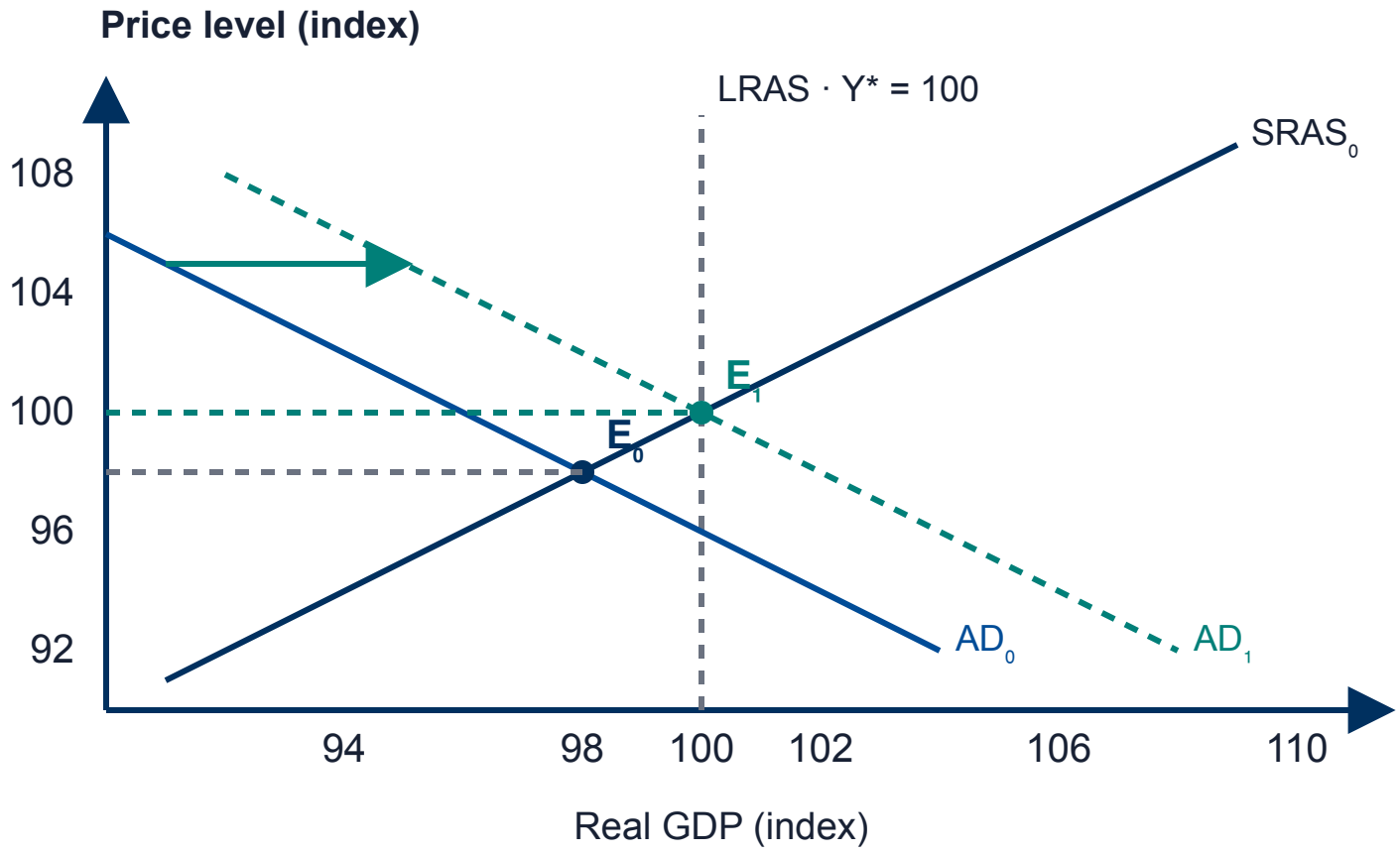
Measuring the Output Gap

Hypothetical case	Actual real GDP	Potential real GDP	Output gap
Demand shortfall	980	1,000	Calculate
Demand pressure	1,020	1,000	Calculate

- Output gap = 100 × (actual GDP − potential GDP) ÷ potential GDP.
- Shortfall: 100 × (980 − 1,000) ÷ 1,000 = −2%.
- Pressure: 100 × (1,020 − 1,000) ÷ 1,000 = +2%.

All GDP inputs are original illustrations in the same real units; potential GDP is estimated.

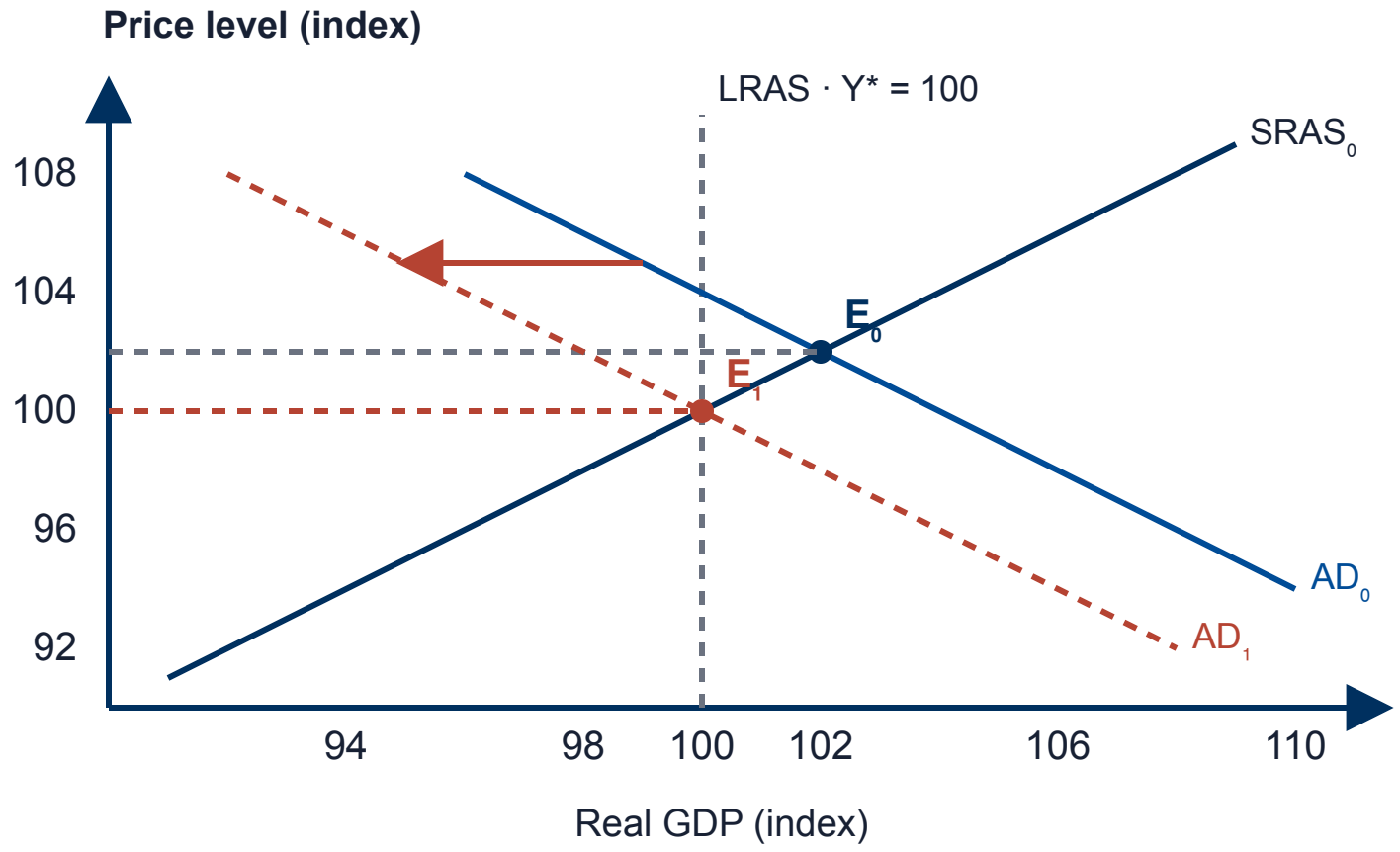
Responding to a Negative Output Gap



Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

1. A negative demand shock leaves output below potential.
2. If inflation pressure is subdued, easing can support C and I.
3. AD moves right toward the fixed potential-output line.
4. Output rises toward potential; the price level rises relative to the weak-demand outcome.

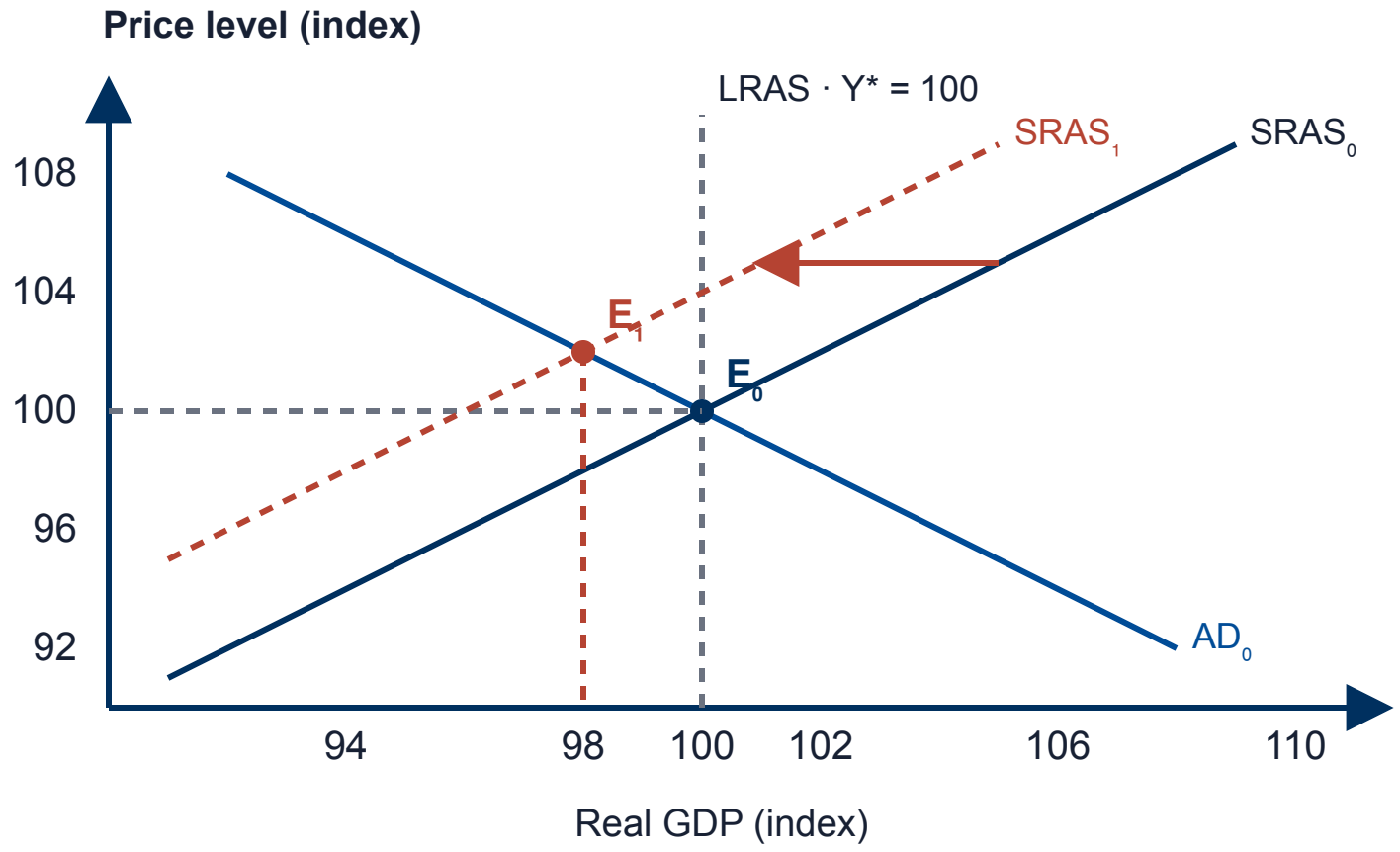
Responding to a Positive Output Gap



Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

1. Strong aggregate demand puts output above sustainable potential.
2. If inflation pressure is excessive, tightening can restrain C and I.
3. AD moves left toward the fixed potential-output line.
4. Output pressure eases, with a lower price level than without tightening.

When Output and Inflation Disagree



Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

1. An adverse supply shock shifts SRAS left/up.
2. Output falls below potential while the price level rises.
3. Easing AD can support output but adds price pressure; tightening AD can reduce price pressure but deepen the output loss.

A Standard Taylor Rule

$$i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5g$$

- i: nominal policy-rate benchmark; r*: neutral real rate.
- π: inflation; π*: inflation goal; g: output gap in percent.
- Higher inflation relative to the goal or a larger output gap raises the benchmark, holding other inputs fixed.

Illustrative 1993-style rule. Rates and gaps enter as percentage numbers, such as 4 and −2.

Federal Reserve Board: policy rules

Calculating a Policy-Rate Benchmark

Fixed illustrative input	Value
Neutral real rate r*	2%
Inflation goal π*	2%
Current inflation π	4%
Output gap g	−2%

- Baseline term: $r^* + \pi = 2 + 4 = 6$.
- Inflation response: $0.5 \times (4 - 2) = +1$.
- Output response: $0.5 \times (-2) = -1$.
- Benchmark: $i = 6 + 1 - 1 = 6\%$.

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Why the Rule Responds to Inflation

> **$\Delta i = 1.5 \times \Delta \pi$ (holding r^*, π^*, and g fixed)**

- One extra percentage point of inflation adds 1 point through π and 0.5 point through the inflation-gap term.
- The nominal rate rises by 1.5 percentage points.
- If expected inflation rises by 1 point too, the approximate real rate rises by 0.5 point.

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Inflation Targeting

- An inflation-targeting framework states an inflation objective and uses forecasts, communication, and policy actions to pursue it over time.
- For the Fed, the longer-run objective is 2% PCE inflation, alongside maximum employment.
- Policy can respond to expected inflation and employment conditions before inflation reaches the goal.

Federal Reserve Board: policy goals and transmission

The September 2024 Decision

Case E · September 18, 2024	Policy fact
Target range change	Cut by 0.50 percentage point
New target range	4.75–5.00%
Statement context	Inflation still somewhat elevated; greater confidence in progress toward 2%

- The committee considered progress on inflation and risks to both employment and inflation.
- A cut before inflation reaches 2% can reflect a forward-looking judgment. The statement does not prove a specific Taylor-rule calculation.

FOMC, September 18, 2024

Taylor Rule and Inflation Targeting

Comparison	Taylor-rule benchmark	Inflation-targeting framework
Main question	What policy rate do these inputs imply?	What outcome should policy pursue over time?
Central inputs	Inflation, inflation gap, output gap, r*	Inflation objective, forecasts, risks, broader mandate
Type of guidance	A numerical instrument guide	An objective and decision framework
Can they coexist?	Yes: a rule can inform the rate decision.	Yes: the target can anchor the rule.

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Why Policymakers Use Judgment

- Potential output and the neutral real rate are estimates that can change.
- Inflation measures and forecasts can disagree; effects of policy arrive with lags.
- The effective lower bound can limit conventional rate cuts. A formula can prescribe an infeasible rate.

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Try the Rule

Fixed input	Value
r* and π*	2% each
Inflation π	3%
Output gap g	−4%

- Calculate the nominal policy-rate benchmark.
- Explain why a negative output gap pushes the benchmark downward.

Explanation. i = 2 + 3 + 0.5(3 − 2) + 0.5(−4) = 3.5%. The output-gap term contributes −2 percentage points.

Write a Conditional Policy Recommendation

Hypothetical economy	Inflation	Output gap	Shock
A	1%	−3%	Demand shortfall
B	5%	−3%	Adverse supply shock

- For each case, recommend a direction and trace the channel through AD.
- Calculate a Taylor benchmark using $r^* = 2\%$, $\pi^* = 2\%$, then explain what the number leaves unresolved.
- Name one piece of evidence that could change your recommendation.

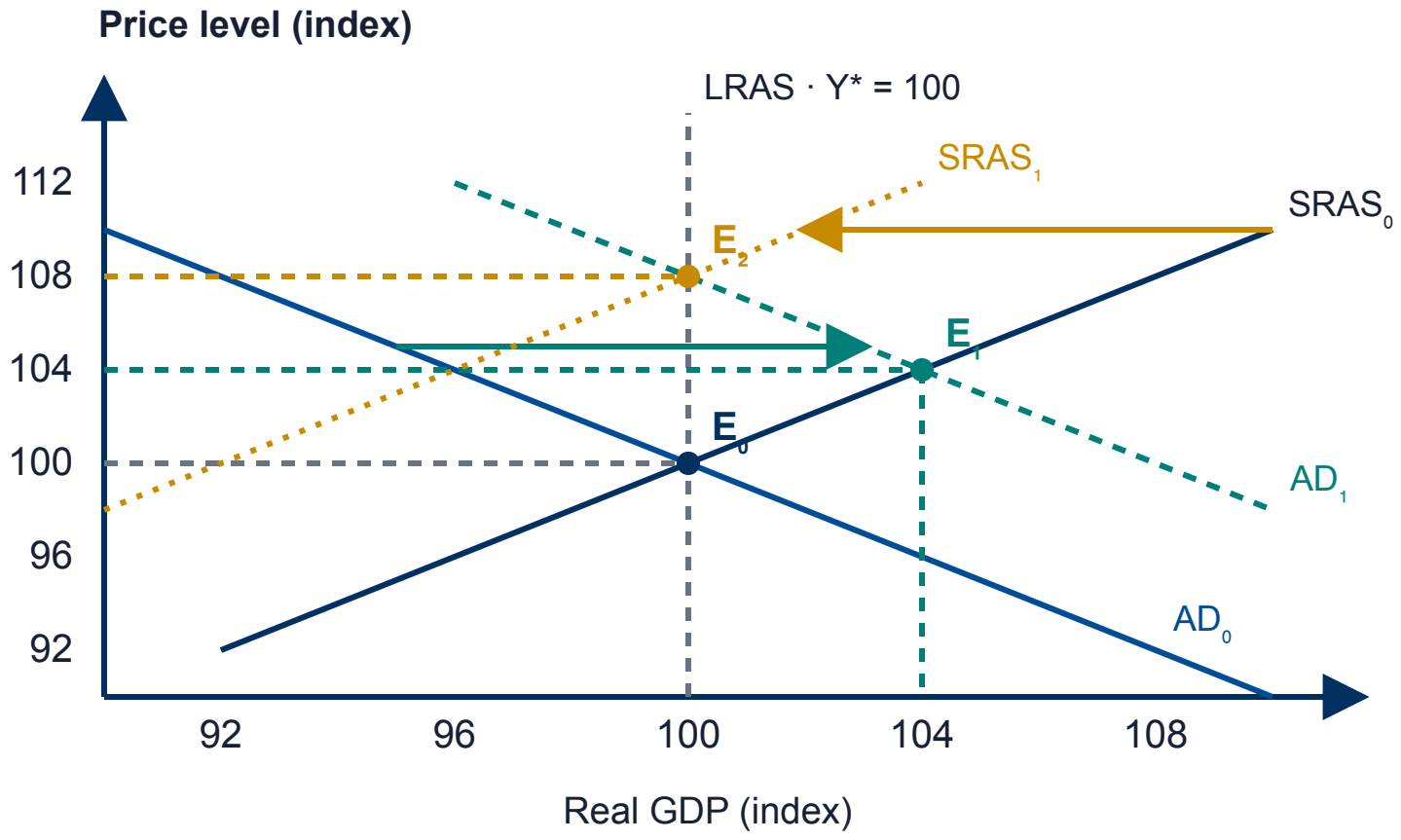
Explanation. A: benchmark 1%; easing can support weak demand if conditions allow. B: benchmark 7%; inflation and output signals conflict, so discuss the tradeoff, expectations, persistence, and other available tools before choosing a response.

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Long-Run Effects of Monetary Policy

- Explain the long-run effects of monetary policy and monetary neutrality: why changes in the money supply affect the price level but not real output in the long run.

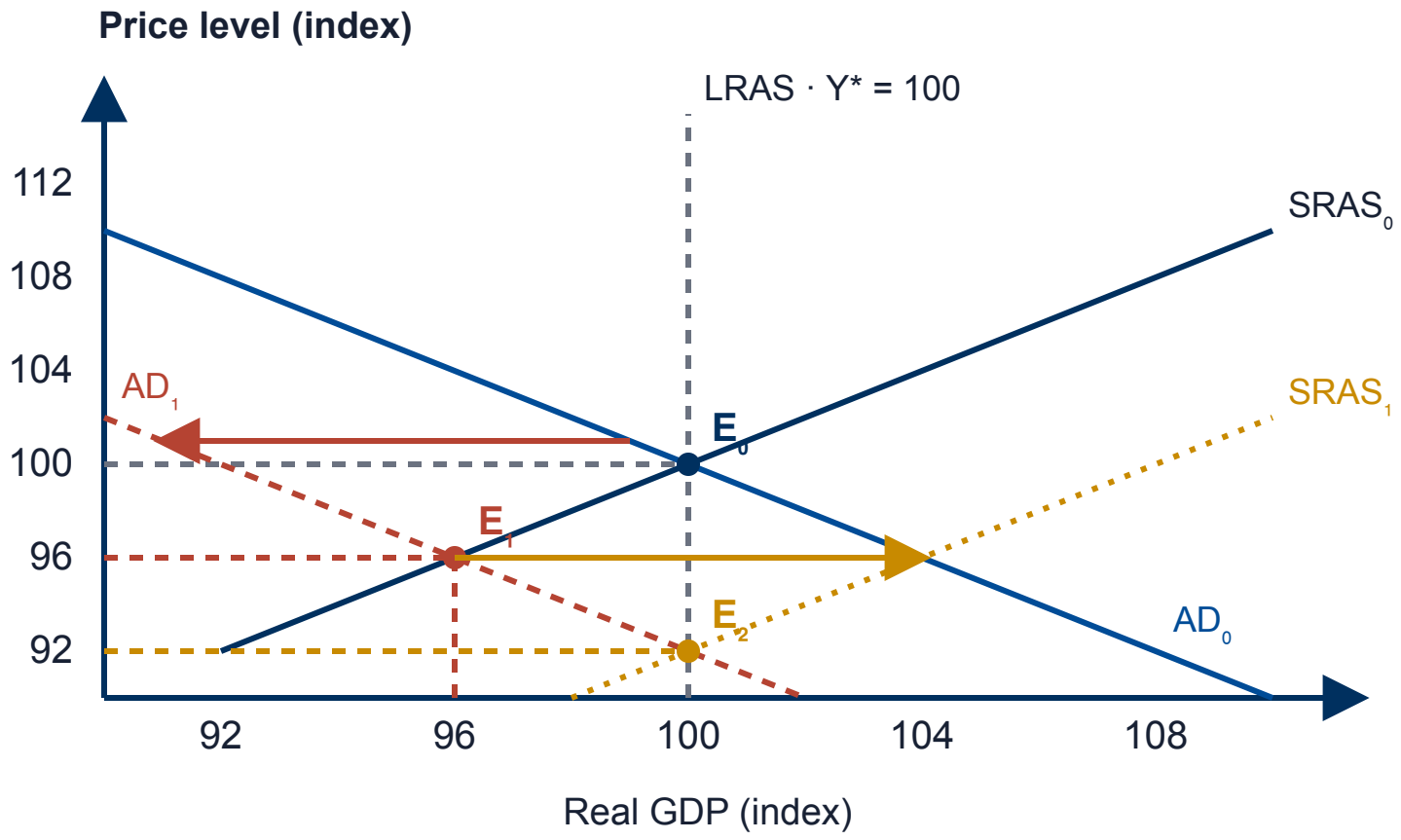
Monetary Expansion: Short Run and Long Run



1. Begin at E_0: output is at potential, $Y^* = 100$, and $P_0 = 100$.
2. A one-time monetary expansion shifts AD_0 right to AD_1.
3. With sticky wages, E_1 has $Y_1 = 104$ and $P_1 = 104$.
4. Above-potential output raises pressure on nominal wages and costs; SRAS shifts left/up.
5. At long-run E_2, output returns to 100; the price level is 108.

Original model: one-time monetary change; wages and prices adjust; productive capacity stays fixed.

Monetary Contraction: Short Run and Long Run



1. Begin at E_0: output is at potential, $Y^* = 100$, and $P_0 = 100$.
2. A one-time monetary contraction shifts AD_0 left to AD_1.
3. With sticky wages, E_1 has $Y_1 = 96$ and $P_1 = 96$.
4. Below-potential output puts downward pressure on nominal wages and costs; SRAS shifts right/down.
5. At long-run E_2, output returns to 100; the price level is 92.

Original model: one-time monetary change; wages and prices adjust; productive capacity stays fixed.

Monetary Neutrality

After a one-time monetary expansion	Short run	Long run
Real output	Can rise above potential	Returns to potential
Price level	Begins to rise	Higher than initially
Nominal wages	Adjust slowly	Rise with prices

- Monetary neutrality: money changes nominal variables without permanently changing real output.
- Potential output depends on labor, capital, productivity, and institutions.

Standard model: one-time monetary change; no change in productive capacity.

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Check Monetary Neutrality

- After a one-time monetary expansion, wages and prices fully adjust. Productive capacity is unchanged. Will real output stay above potential?
- What long-run result would you expect after a one-time monetary contraction under the same conditions?

Explanation. Expansion: real output returns to potential and the price level is higher. Contraction: real output returns to potential and the price level is lower. Short-run output and employment effects can still be substantial.

Policy Reasoning

- Identify the shock and the output gap before recommending a policy.
- Use a rule to organize the evidence; distinguish a benchmark from a binding decision.
- Explain the expected channel, the tradeoff, and the condition that could change your answer.

Explanation. An output gap describes the economy's position. A policy rule supplies a numerical cross-check. An inflation objective and mandate guide the decision over time. Monetary neutrality describes the long-run result after wages and prices adjust.